

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I D.Uday Kiran (192425161) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D.Uday kiran

Scenario 1 – Smart Campus IPv4 Subnetting

1. Introduction

A newly established Smart Campus requires a secure and efficient network infrastructure for different academic departments. The given IPv4 address block is divided into multiple subnets based on the number of devices required by each department.

Subnetting is applied to improve network performance, security, and IP address management by creating separate logical networks.

2. Scenario Understanding

The network administrator has assigned the IP block:

Network Block: 192.168.100.0/24

A /24 network contains:

- Total IP addresses = $2^8 = 256$ addresses
- Usable host addresses = $256 - 2 = 254$ hosts

The network is divided into:

Department	Subnet Mask
School of Computing	/25
School of Electronics	/26
School of Mechanical Engineering	/27

Each department receives a subnet according to its device requirements.

3. Given Data

IPv4 Address Block

192.168.100.0/24

Binary subnet sizes:

Prefix	Total Addresses	Usable Hosts
/25	128	126
/26	64	62
/27	32	30

4. Step-by-Step Subnet Allocation

Subnet 1: School of Computing

Given subnet:

192.168.100.0/25

Subnet mask:

255.255.255.128

Range:

- Network Address:
192.168.100.0
- First Host:
192.168.100.1
- Last Host:
192.168.100.126
- Broadcast Address:
192.168.100.127

Host Calculation

Number of host bits:

$$32 - 25 = 7 \text{ bits}$$

Total addresses:

$$2^7 = 128$$

Usable hosts:

$$128 - 2 = 126 \text{ hosts}$$

Subnet 2: School of Electronics

Next available address:

192.168.100.128

Subnet:

192.168.100.128/26

Subnet mask:

255.255.255.192

Range:

- Network Address:

192.168.100.128

- First Host:
192.168.100.129
- Last Host:
192.168.100.190
- Broadcast Address:
192.168.100.191

Host Calculation

Number of host bits:

$$32 - 26 = 6 \text{ bits}$$

Total addresses:

$$2^6 = 64$$

Usable hosts:

$$64 - 2 = 62 \text{ hosts}$$

Subnet 3: School of Mechanical Engineering

Next available address:

192.168.100.192

Subnet:

192.168.100.192/27

Subnet mask:

255.255.255.224

Range:

- Network Address:
192.168.100.192
- First Host:
192.168.100.193
- Last Host:
192.168.100.222
- Broadcast Address:
192.168.100.223

Host Calculation

Number of host bits:

$$32 - 27 = 5 \text{ bits}$$

Total addresses:

$$2^5 = 32$$

Usable hosts:

$$32 - 2 = 30 \text{ hosts}$$

5. Remaining Unused IP Address Range

Total network:

192.168.100.0/24

Allocated:

Computing

192.168.100.0 – 192.168.100.127

Electronics

192.168.100.128 – 192.168.100.191

Mechanical

192.168.100.192 – 192.168.100.223

Remaining addresses:

192.168.100.224 – 192.168.100.255

This range contains:

Total addresses:

32

Usable hosts:

30

6. Result Table

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
School	192.168.100	192.168.100	192.168.100	126

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
School of Computing	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	62
School of Electronics	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

7. Decision Making

The subnet sizes are selected according to department requirements:

- The School of Computing receives the largest subnet (/25) because it requires more devices such as computers, servers, and lab systems.
- The School of Electronics receives a medium subnet (/26).
- The School of Mechanical Engineering receives a smaller subnet (/27).

This allocation avoids IP wastage and improves network scalability.

8. Applications

The subnet design can be applied in:

- Smart campus network management
- Department-wise access control
- Secure communication between departments
- Efficient IP address utilization
- Future network expansion

Scenario 2 – Company IPv4 Subnetting

1. Introduction

A multinational software company is expanding its headquarters network infrastructure. To support different departments with different numbers of employees, the organization uses subnetting to divide a large private IP address block into smaller logical networks. Subnetting improves network performance, security, scalability, and efficient IP address utilization.

2. Scenario Understanding

The company has been allocated the private IP address block:

Network Block: 10.10.0.0/16

A /16 network contains:

- Total IP addresses = $2^{16} = 65,536$ addresses
- Usable host addresses = **65,534 hosts**

Different departments require different subnet sizes:

Department	Required Subnet
Software Development	/24
Quality Assurance	/25
Human Resources	/26
Executive Management	/27

The subnet allocation is performed using Variable Length Subnet Masking (VLSM).

3. Given Data

Prefix Length	Subnet Mask	Total Addresses	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30

4. Step-by-Step Solution

Subnet 1: Software Development Department

Allocated subnet:

10.10.0.0/24

Subnet Mask:

255.255.255.0

Address Details

- Network Address:

10.10.0.0

- First Host:

10.10.0.1

- Last Host:

10.10.0.254

- Broadcast Address:

10.10.0.255

Host Calculation

Host bits:

$$32 - 24 = 8 \text{ bits}$$

Total addresses:

$$2^8 = 256$$

Usable hosts:

$$256 - 2 = \mathbf{254 \text{ hosts}}$$

Subnet 2: Quality Assurance Department

Next available address:

10.10.1.0

Allocated subnet:

10.10.1.0/25

Subnet Mask:

255.255.255.128

Address Details

- Network Address:

10.10.1.0

- First Host:

10.10.1.1

- Last Host:

10.10.1.126

- Broadcast Address:

10.10.1.127

Host Calculation

Host bits:

$$32 - 25 = 7 \text{ bits}$$

Total addresses:

$$2^7 = 128$$

Usable hosts:

$$128 - 2 = \mathbf{126 \text{ hosts}}$$

Subnet 3: Human Resources Department

Next available address:

10.10.1.128

Allocated subnet:

10.10.1.128/26

Subnet Mask:

255.255.255.192

Address Details

- Network Address:

10.10.1.128

- First Host:

10.10.1.129

- Last Host:

10.10.1.190

- Broadcast Address:

10.10.1.191

Host Calculation

Host bits:

$$32 - 26 = 6 \text{ bits}$$

Total addresses:

$$2^6 = 64$$

Usable hosts:

$$64 - 2 = \mathbf{62 \text{ hosts}}$$

Subnet 4: Executive Management Department

Next available address:

10.10.1.192

Allocated subnet:

10.10.1.192/27

Subnet Mask:

255.255.255.224

Address Details

- Network Address:

10.10.1.192

- First Host:

10.10.1.193

- Last Host:

10.10.1.222

- Broadcast Address:

10.10.1.223

Host Calculation

Host bits:

$$32 - 27 = 5 \text{ bits}$$

Total addresses:

$$2^5 = 32$$

Usable hosts:

$$32 - 2 = \mathbf{30 \text{ hosts}}$$

5. Unused IP Address Range

Original network:

10.10.0.0/16

Allocated ranges:

Software Development

10.10.0.0 – 10.10.0.255

Quality Assurance

10.10.1.0 – 10.10.1.127

Human Resources

10.10.1.128 – 10.10.1.191

Executive Management

10.10.1.192 – 10.10.1.223

Remaining unused range:

10.10.1.224 – 10.10.255.255

This remaining address space can be used for:

- Future departments
- Additional servers
- Expansion of existing networks

6. Result Table

Departm ent	Networ k Addres s	Valid Host Range	Broadc ast Addres s	Usa ble Hos ts
Software Develop ment	10.10.0. 0	10.10.0. 1 – 10.10.0.	10.10.0. 255	254

Department	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
		254		
Quality Assurance	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

7. Decision Making: The subnet allocation is performed according to department size:

- Software Development receives the largest subnet (/24) because it contains more developers, systems, and testing environments.
- Quality Assurance receives a medium subnet (/25) for testing systems.
- Human Resources receives a smaller subnet (/26).
- Executive Management receives the smallest subnet (/27) because fewer devices are required.

VLSM ensures maximum IP utilization and avoids unnecessary address wastage.

8. Applications

This subnet design can be applied in:

- Enterprise network architecture
- Department-wise network isolation
- Secure data communication
- Employee device management
- Future company expansion

Scenario 3 – IPv6 Addressing

1. Introduction

A smart hospital is upgrading its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices, IoT sensors, smart monitoring systems, and advanced healthcare applications.

IPv6 provides a much larger address space, improved security features, and efficient routing compared to IPv4.

The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

2. Scenario Understanding

IPv6 addresses are 128-bit hexadecimal addresses represented using eight groups separated by colons.

Format:

XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX

The given prefix length is:

/64

Meaning:

- First 64 bits → Network Prefix
- Remaining 64 bits → Host Interface ID

A /64 IPv6 subnet provides a very large number of host addresses for IoT and medical devices.

3. Given Data

Given IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000/64

Number of bits:

128 bits

Network portion:

64 bits

Host portion:

64 bits

4. Step-by-Step Solution

1. IPv6 Address Compression

IPv6 compression rules:

1. Remove leading zeros from each block.
2. Replace continuous groups of zeros with (::).

Given address:

2001:0db8:abcd:0000:0000:0000:0000:0000

Remove leading zeros:

2001:db8:abcd:0:0:0:0:0

Replace consecutive zeros using (::):

Compressed IPv6 Address

2001:db8:abcd::/64

2. Expansion of Compressed Address

Given compressed address:

2001:db8:abcd::/64

The missing groups are filled with zeros.

Expanded form:

2001:0db8:abcd:0000:0000:0000:0000:0000

Full Hexadecimal Representation:

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Identification of Network Prefix

Prefix length:

/64

The first four groups represent the network portion:

2001:0db8:abcd:0000

Therefore,

Network Prefix:

2001:0db8:abcd:0000::/64

or

2001:db8:abcd::/64

The remaining four groups are used for host identification.

4. Calculation of Total Host Addresses

IPv6 address length:

= 128 bits

Network bits:

= 64 bits

Host bits:

= 128 - 64

= **64 bits**

Number of host addresses:

$$2^{64} \\ = 18,446,744,073,709,551,616$$

Total Host Addresses:

18,446,744,073,709,551,616 addresses

5. Total Possible Host Addresses Available

For a /64 IPv6 subnet:

Host portion:

0000:0000:0000:0000

-

ffff:ffff:ffff:ffff

Therefore:

$$2^{64}$$

Possible addresses are available.

Available Hosts:

18.4 Quintillion IPv6 addresses

6. Result Table

Parameter	Result
Original IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Compressed Address	2001:db8:abcd::/64
Expanded Address	2001:0db8:abcd:0000:0000:0000:0000:0000
Network Prefix	2001:0db8:abcd:0000::/64
Network Bits	64 bits
Host Bits	64 bits
Total Host Addresses	2^{64}
Available Addresses	18,446,744,073,709,551,616

7. Decision Making

IPv6 /64 addressing is selected because:

- It provides a huge address space for medical IoT devices.
- Every device can have a unique global address.
- It reduces the need for Network Address Translation (NAT).
- It supports future hospital expansion.
- It improves routing efficiency and security.

8. Applications

IPv6 implementation in smart hospitals supports:

- Smart patient monitoring devices
- IoT healthcare sensors
- Medical equipment communication
- Remote healthcare systems
- Automated hospital management systems

Scenario 4 – Routing Table Analysis

1. Introduction

An Internet Service Provider (ISP) connects multiple customer networks using a core router. The router maintains a routing table that contains information about different

destination networks and determines the best path for forwarding incoming packets.

Routing tables help routers make forwarding decisions by matching the destination IP address with available network entries.

2. Scenario Understanding

The ISP core router has the following routing table entries:

Routing Entry	Network
1	192.168.1.0/24
2	192.168.2.0/24
3	192.168.3.0/24

A packet arrives with destination IP:

192.168.2.45

The router must identify the correct subnet, routing entry, forwarding interface, and default route.

3. Given Data

Routing Table

Network Address	Prefix Length	Subnet Mask
192.168.1.0	/24	255.255.255.0
192.168.2.0	/24	255.255.255.0
192.168.3.0	/24	255.255.255.0

Destination IP:

192.168.2.45

4. Step-by-Step Solution

1. Determine Destination Subnet

Given destination address:

192.168.2.45

Subnet mask:

/24

Binary representation:

Network Portion: 192.168.2

Host Portion: 45

The first three octets represent the network.

Therefore:

Destination Network:

192.168.2.0/24

2. Matching Routing Table Entry

The router compares the destination IP with all available routes.

Routing Table Comparison:

Routing Entry	Match Result
192.168.1.0/24	No Match
192.168.2.0/24	Match Found
192.168.3.0/24	No Match

The destination IP belongs to:

Matching Route:

192.168.2.0/24

3. Next-Hop Network/Interface Used

Since the destination network exists directly in the routing table, the router forwards the packet through the interface connected to:

192.168.2.0/24 network

The forwarding decision:

Incoming Packet

|

↓

Core Router

|

↓

Interface connected to 192.168.2.0/24



Destination Host 192.168.2.45

Next-Hop Interface:

192.168.2.0/24 network interface

4. Default Route for Unknown Destinations

If the destination IP does not match any routing table entry, the router uses a default route.

Default route notation:

0.0.0.0/0

Meaning:

- Matches all unknown destinations.
- Sends packets to an upstream ISP router.

Default Route:

0.0.0.0/0

5. Result Table

Parameter	Result
Destination IP	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Routing Entry	192.168.2.0/24
Forwarding Interface	Interface connected to 192.168.2.0 network
Default Route	0.0.0.0/0

6. Decision Making

The router uses the **Longest Prefix Match algorithm** to select the best route.

In this scenario:

- The destination IP matches the 192.168.2.0/24 entry.
- The router forwards the packet directly through the corresponding interface.
- No default route is required because a specific route is available.

If no matching route exists, the router forwards the packet using the default route.

7. Applications

Routing table analysis is used in:

- Internet Service Provider networks
- Enterprise routers
- Data center communication
- Packet forwarding decisions
- Network troubleshooting

Scenario 5 – LAN & Router Configuration

1. Introduction

A company has established two separate office departments connected through a Layer-3 router. The router enables communication between different LAN networks by performing routing functions.

Subnetting is used to divide the available IP address space into separate networks for the Administration and Finance departments. Proper gateway configuration allows devices in both LANs to communicate efficiently.

2. Scenario Understanding

The company has assigned the following subnet information:

Department	Network Address	Prefix
Administration LAN	192.168.10.0	/26
Finance LAN	192.168.10.64	/26

Router interface configuration:

Department	Default Gateway
Administration LAN	192.168.10.1
Finance LAN	192.168.10.65

Each LAN requires:

- Network address identification
- Broadcast address calculation
- Valid host range identification

- Computer IP assignment
- Default gateway configuration

3. Given Data

Subnet Mask for /26:

255.255.255.192

Host bits:

$$32 - 26 = 6 \text{ bits}$$

Total addresses:

$$2^6 = 64$$

Usable hosts:

$$64 - 2 = 62$$

Each LAN can support:

62 usable devices

4. Step-by-Step Solution

LAN 1: Administration Department

Given Network:

192.168.10.0/26

Subnet Mask:

255.255.255.192

Address Range Calculation

Block size:

$$256 - 192 = 64$$

Therefore, first subnet range:

192.168.10.0 – 192.168.10.63

Network Address:

192.168.10.0

First Host Address:

192.168.10.1

Last Host Address:

192.168.10.62

Broadcast Address:

192.168.10.63

Suggested Computer IP Addresses

Device	IP Address
Computer 1	192.168.10.10
Computer 2	192.168.10.11
Computer 3	192.168.10.12

Default Gateway:

192.168.10.1

LAN 2: Finance Department

Given Network:

192.168.10.64/26

Subnet Mask:

255.255.255.192

Address Range Calculation

Second subnet range:

192.168.10.64 – 192.168.10.127

Network Address:

192.168.10.64

First Host Address:

192.168.10.65

Last Host Address:

192.168.10.126

Broadcast Address:

192.168.10.127

Suggested Computer IP Addresses

Device	IP Address
Computer 1	192.168.10.70
Computer 2	192.168.10.71
Computer 3	192.168.10.72

Default Gateway:

192.168.10.65

5. Result Table

LAN	Network Address	Valid Host Range	Broadcast Address	Default Gateway
Administration LAN	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63	192.168.10.1
Finance LAN	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127	192.168.10.65

6. Computer IP Allocation Table

Department	Computer	Assigned IP Address	Gateway
Administration	PC 1	192.168.10.10	192.168.10.1
Administration	PC 2	192.168.10.11	192.168.10.1
Administration	PC 3	192.168.10.12	192.168.10.1

Department	Computer	Assigned IP Address	Gateway
Finance	PC 1	192.168.10.70	192.168.10.65
Finance	PC 2	192.168.10.71	192.168.10.65
Finance	PC 3	192.168.10.72	192.168.10.65

7. Decision Making

The subnet allocation decision is based on efficient network design:

- Both departments receive equal-sized /26 subnets because they require similar numbers of devices.
- The router interfaces are assigned the first usable IP address of each subnet as the default gateway.
- Separate subnet ranges improve security and reduce broadcast traffic.
- Layer-3 routing enables communication between Administration and Finance networks.

8. Applications

This LAN design is useful for:

- Enterprise office networks
- Department-wise network segregation
- Secure internal communication
- Router-based interdepartmental connectivity
- Efficient IP address management